

T13 — X-ray data-quality oracle (new in this re-issue)

scripts/t13_data_quality.py · CCP4 ctruncate (with aimless attempted) · closes the last cross-tool gap

Metric	Tool	Value	Verdict
Wilson B	ctruncate	14.54 Å ² (σ 21.12)	informational · matches refined protein-mean B 16.0
L-test twin fraction	ctruncate	0.03	✓ < 0.05 — untwinned
Moments-based twin	ctruncate	0.02	✓ untwinned (<I ² >/<I> ² = 1.997, expected 2.0)
L statistic	ctruncate	0.505	✓ untwinned (untwinned 0.500, perfect twin 0.375)
First-principles twin operators	ctruncate	0	✓ none found by lattice / symmetry search
ΔB anisotropy	ctruncate	7.32 Å ² (eigvals 10.56 / 17.88 / 14.63)	✓ < 20 Å ² rule of thumb · mild but present
tNCS (Patterson 4 Å)	ctruncate	false	✓ no off-origin peaks > 14 Å
Ice rings	ctruncate	1 flagged at 3.44 Å (Z 5.21, completeness 1.00)	☐ informational · borderline · not a refinement blocker
aimless (canonical T13)	CCP4 aimless	aborted	☐ MTZ has merged amplitudes only (no M/ISYM)

Coverage now closed — every catalog task green

T13 was the only "partial" entry in the cross-tool matrix at 2026-05-01.
ctruncate (CCP4 / non-cctbx) supplies 5 metrics computable from merged F-obs.
Five findings now have ≥ 3 confirmations (R-free · Asn A 39 · RMSD bracket ·
ligand RSCC · Mg²⁺ exclusion); data quality has 1 cctbx + 1 non-cctbx.

Wrapper persists logs at data/coscientists/openscientist/t13_oracle_logs/

aimless limitation — captured as provenance

aimless requires unmerged intensities (M/ISYM column).
1sar.mtz ships only F-obs / SIGF-obs → aimless aborts:
hkl_unmerge_list::prepare – EMPTY
CC½, ⟨I/σ⟩ outer, Rmerge / Rmeas remain unobtainable —
data-availability gap, not an oracle failure.
Documented in skill `Tool assumptions` for future artefacts.